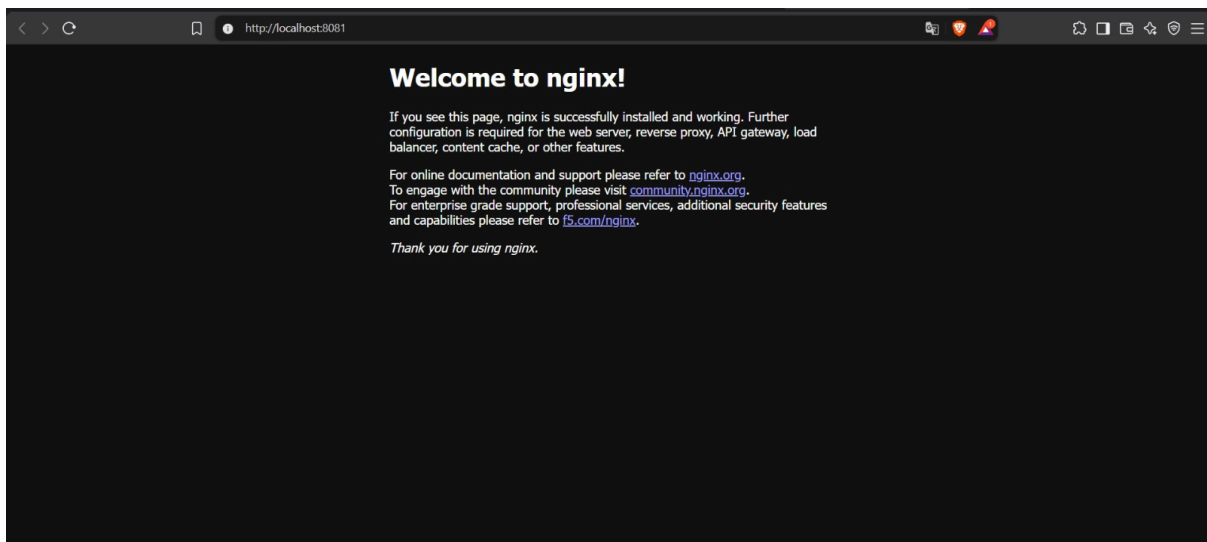


EJERCICIOS DIAPOSITIVAS

1.imágenes

```
PS C:\taller-docker-git\3Agosto> docker run -d --name practica1 -p 8081:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
f340c1b7c1d6: Pull complete
07db7bf2649b: Pull complete
a44b5c8be616: Pull complete
c12f394dea35: Pull complete
PS C:\taller-docker-git\3Agosto> docker ps          # contenedores corriendo
>> docker stop practica1
>> docker rm practica1
>> docker ps -a          # debería quedar vacío (o sin practica1) al final
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
91613cbb1c69   nginx         "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp  practica1
practica1
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
6aee3ff24093   gcr.io/k8s-minikube/kicbase:v0.0.50  "/usr/local/bin/entr..." 7 days ago     Exited (137) 2 days ago                                minikube
PS C:\taller-docker-git\3Agosto>
```



2. Primer y segundo plano

```
PS C:\taller-docker-git\3Agosto> docker run -it --name consola ubuntu:24.04 bash
Unable to find image 'ubuntu:24.04' locally
24.04: Pulling from library/ubuntu
0926a8eb0e60: Pull complete
989d374f715e: Download complete
Digest: sha256:33ceb71981b602c1a7443a53469e4dba065f7503eab3078a2d7a57a2ab987517
Status: Downloaded newer image for ubuntu:24.04
root@54eff5942fc4:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@54eff5942fc4:/# ^C
root@54eff5942fc4:/# exit
```

```

What's next:
  Debug this container error with Gordon →docker at "help me fix this container error"
PS C:\taller-docker-git\3Agosto> docker run -d --name web -p 8080:80 nginx
f1edc8c4c9d35e019ab302a4e04006078097220902282dfa087662086162c89
PS C:\taller-docker-git\3Agosto> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
f1edc8c4c9d3   nginx    "/docker-entrypoint. ..." 10 seconds ago Up 9 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web
PS C:\taller-docker-git\3Agosto> docker logs -f web
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/03 18:02:53 [notice] 1#1: using the "epoll" event method
2026/09/03 18:02:53 [notice] 1#1: nginx/1.31.5
2026/09/03 18:02:53 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/03 18:02:53 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/09/03 18:02:53 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576

```

3. Docker Exec

```

PS C:\taller-docker-git\3Agosto> docker exec -it web bash
root@f1edc8c4c9d3:/# ^C
root@f1edc8c4c9d3:/# ls /usr/share/nginx/html
50x.html  index.html
root@f1edc8c4c9d3:/#

```

4. Variables de Entorno

```

PS C:\taller-docker-git\3Agosto> docker run -d --name db -e POSTGRES_USER=admin -e POSTGRES_PASSWORD=secreto -e POSTGRES_DB=tienda postgres:16
44a09da885f54578a238835489c987670fbcb9005d5fb26d8352f1becff54fd9
PS C:\taller-docker-git\3Agosto> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
44a09da885f5   postgres:16 "docker-entrypoint.s ..." 6 seconds ago Up 6 seconds   5432/tcp                               db
f1edc8c4c9d3   nginx    "/docker-entrypoint. ..." 14 minutes ago Up 14 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web
PS C:\taller-docker-git\3Agosto> docker exec db env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/lib/postgresql/16/bin
HOSTNAME=44a09da885f5
POSTGRES_PASSWORD=secreto
POSTGRES_DB=tienda
POSTGRES_USER=admin
GOSU_VERSION=1.19
LANG=en_US.utf8
PG_MAJOR=16
PG_VERSION=16.14-1.pgdg13+1

```

5. volumenes

```

> docker run -d --name db -v datos_db:/var/lib/postgresql/data -e POSTGRES_PASSWORD=x postgres:16
atos_db
c0ebc24980dc457a12e95b356f5ce2738014769c610eb63f399a95772b3fc38
PS C:\taller-docker-git\3Agosto> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
c0ebc24980dc   postgres:16 "docker-entrypoint.s ..." 6 seconds ago Up 6 seconds   5432/tcp                               db
cd027c064c9    nginx    "/docker-entrypoint. ..." 17 minutes ago Up 17 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   api
f1edc8c4c9d3   nginx    "/docker-entrypoint. ..." 33 minutes ago Up 33 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web
RIVER         VOLUME NAME
ocal         0baa52b6869f610ca7f057bb754710425da5ba5725baf27b73e7b03d0a1f6d5e
ocal         274d514c096544af9e7f804bc427c631e2144d48cf8597bcb43bc905f19dab9d
ocal         455fd7940ef9f27c2cb0ab7132b95805a23debf1bf3d509a94179e6739a48361
ocal         580dcacc5f7ae01f8eba3474d0a05e7c62da909809503100764e1e7ac70b23d
ocal         c0ba601b989ebf7fd75d73ec9f16b05af2543155874ebf26f2ea2e60b82e9247
ocal         datos_db
ocal         ejemplo_db_data

```

6. redes

```
PS C:\taller-docker-git\3Agosto> docker network create mired
0db6a73603879cddb030c1d6c33490fd39d4f4aaa3ac051f98cfce27e7f2b3d3
PS C:\taller-docker-git\3Agosto> docker run -d --name servidor --network mired nginx
bad0c41634ab46297297858402a3fb046d6059f8fbc41f68b4fde4a0de784bbd
PS C:\taller-docker-git\3Agosto> docker run --rm -it --network mired busybox ping -c 2 servidor
PING servidor (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.502 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.127 ms

--- servidor ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.127/0.314/0.502 ms
```

7. practica 2

```
What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image →docker debug db
  Learn more at https://docs.docker.com/go/debug-cli/
PS C:\taller-docker-git\3Agosto> docker stop db
db
PS C:\taller-docker-git\3Agosto> docker rm db
db
PS C:\taller-docker-git\3Agosto> docker run -d --name db --network red_demo -v vol_demo:/var/lib/postgresql/data -e POSTGRES_PASSWORD=demo postgres:16
67529958ee8d3a34c402a550f2d0ac57cc2b2b46931790409a4c650c90909aa6
PS C:\taller-docker-git\3Agosto> docker exec -it db psql -U postgres -c "SELECT * FROM notas;"
t
-----
sobrevivi
(1 row)
```

8. Dockerfile

```
PS C:\taller-docker-git\app-03> curl http://localhost:5000

Advertencia de seguridad: riesgo de ejecución de script
Invoke-WebRequest analiza el contenido de la página web. El código de script de la página web se puede ejecutar cuando se analiza la página.
ACCIÓN RECOMENDADA:
Usa el modificador -UseBasicParsing para evitar la ejecución de código de script.

¿Quieres continuar?

[S] Sí [O] Sí a todo [N] No [T] No a todo [U] Suspender [?] Ayuda (el valor predeterminado es "N"): S

StatusCode      : 200
StatusDescription: OK
Content          : Hola desde mi contenedor
RawContent       : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 24
                  Content-Type: text/html; charset=utf-8
                  Date: Thu, 03 Sep 2026 20:19:16 GMT
                  Server: Werkzeug/3.1.8 Python/3.12.14
```

9.ejercicio

```
#10 exporting manifest list sha256:196efe310079635a44ac39b228661afa451d3c0a5850c215bb54a3aaa80df39d 0.0s done
#10 naming to docker.io/library/miniblog-web:latest done
#10 unpacking to docker.io/library/miniblog-web:latest
#10 unpacking to docker.io/library/miniblog-web:latest 0.3s done
#10 DONE 1.5s

#11 resolving provenance for metadata file
#11 DONE 0.0s
[+] up 5/5
✓ Image miniblog-web Built 6.7s
✓ Network miniblog_default Created 0.0s
✓ Volume miniblog_datos_miniblog Created 0.0s
✓ Container miniblog-db-1 Started 0.5s
✓ Container miniblog-web-1 Started 0.6s
PS C:\taller-docker-git\miniblog> docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
miniblog-db-1	postgres:16	"docker-entrypoint.s..."	db	11 seconds ago	Up 10 seconds	5432/tcp
miniblog-web-1	miniblog-web	"python app.py"	web	11 seconds ago	Up 10 seconds	0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp

```
PS C:\taller-docker-git\miniblog> curl.exe http://localhost:8080
MiniBlog funcionando, conectado a la base de datos
PS C:\taller-docker-git\miniblog>
```

10. CI y CD

The top screenshot shows the GitHub Actions interface for the repository `ismaelmira-dev / taller-docker-git`. The `Actions` tab is selected, displaying a list of workflow runs. Two runs are visible:

- Corrige app.run y test con status_code correcto** (Workflow #2, Commit 5518cc7, pushed by ismaelmira-dev, 3 Septiembre, 7 minutes ago, 16s).
- Agrega workflow CI con GitHub Actions (Modulo 5)** (Workflow #1, Commit 01e2dce, pushed by ismaelmira-dev, 3 Septiembre, 16 minutes ago, 15s).

The bottom screenshot shows a VS Code editor with the `test_app.py` file open. The file contains the following Python code:

```
test_app.py > test_inicio
You, hace 7 minutos | 1 author (You)
1 from app import app
2
3 def test_inicio():
4     cliente = app.test_client()
5     r = cliente.get("/")
6     assert r.status_code == 200
```